Simple Predictions of Uniform-Start Random-Walk Cover Times on Small Graphs: Accuracy and Family Dependence

Autonomous Research Agent [1]

Abstract

Can a few inexpensive graph statistics predict random-walk cover times more closely than a general upper bound? We generate 133 connected graphs with 8–30 vertices from ten families and estimate their uniform-start mean cover times with 4,000 independent walks each. A fixed log-linear ridge model using vertex count, edge count, maximum degree, and diameter achieves a family-balanced test mean absolute percentage error of 20.1%, with a Monte Carlo 95% interval of [19.8, 20.5]%, compared with 75.9% for a fitted size-only model. Accuracy depends strongly on family: errors are about 5% on stars but 50–55% on grids and dense random graphs. Leaving an entire family out of training raises overall error to 31.3%. These results support a useful empirical approximation on this graph collection, without a uniform accuracy guarantee over small connected graphs.

1. Introduction

For a simple random walk on a finite connected undirected graph, the time to visit all vertices varies substantially with topology. General bounds, including the classical edge-based bound associated with random-walk traversal (Aleliunas et al., 1979), give guarantees but need not approximate individual instances. Tight extremal bounds (Feige, 1995) address a different objective from typical-instance prediction. The survey of Lovász (1993) provides background on random walks on graphs.

We test four basic statistics on held-out sizes and unseen families, with saved raw trajectories and propagated simulation uncertainty. Under the task's access restriction, no research literature or discussing pages were retrieved; we make no novelty claim. The supplied references serve as background; all numerical evidence was generated here.

[1]Independent reproducible experiment. Correspondence to: reproduction instructions in the accompanying README.

Independent experimental report, October 2026. Official ICML 2025 format; not a conference submission or publication.

2. Method

Target and estimator. Let $G = (V, E)$, $n = |V|$, $m = |E|$, and let X_0 be uniform on V. At each step, the walk chooses a neighbor uniformly. We study

$$C_U(G) = \mathbb{E}\big[\min\{t : \{X_0, \dots, X_t\} = V\}\big]. \quad (1)$$

Thus the initial vertex is visited at time zero. This target averages over starting vertices; it is not the frequently used worst-start quantity $\max_v \mathbb{E}_v[T]$. For each graph we simulate $R = 4000$ independent walks, independently resampling the initial vertex each time, with no trajectory cutoff. The label $\widehat{C}_U$ is their sample mean; its standard error is the sample standard deviation divided by $\sqrt{R}$. We save every integer cover time and report a Student-t approximate 95% interval for each graph mean.

Predictors. Our four features are $x = (\log n, \log m, \log \Delta, \log D)$, where Δ is maximum degree and D is diameter. Average degree $2m/n$ adds no independent information. Breadth-first searches compute D in $O(n(n+m))$ time, negligible here. Feature means and standard deviations are computed on training graphs only. Writing standardized features as z, we fit $\log \widehat{C}_U$ by minimizing

$$\sum_{g \in \mathcal{T}} w_g (\log \widehat{C}_U(g) - b - \beta^\top z_g)^2 + 0.01\|\beta\|_2^2, \quad (2)$$

where weights sum to one, every family has equal total weight, and graphs within each family have equal weight. The intercept is unpenalized. Predictions are $\exp(b + \beta^\top z)$, with no correction or clipping. The feature set, penalty, and splits were fixed before inspecting scores; we perform no hyperparameter search.

We compare with the same fitted procedure using only $\log n$, and with $B(G) = 2m(n-1)$, a valid general upper bound on worst-start cover time and hence on C_U. A bound is a deliberately conservative reference, not a competitive learned predictor. We do not use an asymptotic extremal formula as a finite-n guarantee.

Evaluation and uncertainty. For a set of graphs, define MAPE as the family-balanced average of $100|\widehat{C}_{\mathrm{pred}}/\widehat{C}_U - 1|$. We also report the multiplicative error factor $\exp(\mathrm{avg}\,|\log(\widehat{C}_{\mathrm{pred}}/\widehat{C}_U)|)$ using the same weights. A factor of one is perfect.

To propagate Monte Carlo noise through both fitting and evaluation, we partition each graph's 4,000 independent times into 40 blocks of 100. In each of 500 bootstrap repetitions we independently resample the 40 block means for every graph, reconstruct its mean, refit every model, and recompute scores. Brackets in the tables are the 2.5th and 97.5th percentiles. These are approximate Monte Carlo intervals conditional on the fixed graph collection, not uncertainty over unseen graphs, a prediction interval, or a guarantee of model accuracy. Graphs, rather than individual trajectories, are the units of prediction.

3. Experiments

Graph collection and splits. We use $n \in \{8, 12, 16, 20, 24, 28, 30\}$. Seven deterministic families have one graph per size: path, cycle, star, clique, grid, lollipop, and barbell. The grid comprises vertices $0, \ldots, n-1$ arranged in rows of width $\lfloor\sqrt{n}\rfloor$, with horizontal and vertical lattice edges; the final row may be incomplete. The lollipop has a clique of $\lceil n/2\rceil$ vertices attached to a path on the remaining vertices. The barbell has two cliques of size $\lfloor n/3\rfloor$, joined through a path with $n - 2\lfloor n/3\rfloor$ internal vertices. Three stochastic families have four independent graphs per size: $G(n, 3/(n-1))$, $G(n, 0.35)$, and uniform labeled trees generated from uniform Prüfer sequences. ER graphs are rejection-sampled until connected. The sparse/dense names refer to the parameter rules, not to an ordering of realized densities at every size.

Training sizes are $\{8, 12, 20, 28\}$ (76 graphs); testing sizes are $\{16, 24, 30\}$ (57 graphs), including a small extrapolation to 30. No deterministic graph is duplicated across splits. A separate leave-one-family-out (LOFO) evaluation fits on all sizes of nine families and tests all sizes of the remaining family, repeated ten times. LOFO and held-out-size results therefore answer different questions. The global seed is 20261027; graph-specific seeds and edge lists are stored. The complete CPU experiment takes under one minute, including compilation, simulation, fitting, and 500 bootstraps.

Simulation checks. For every sampled size we compare against three analytic uniform-start expectations. With $L = n-1$ and $H_L = \sum_{k=1}^{L} 1/k$, these are $L^2 + L(L-1)/6$ for a path, LH_L for a clique, and $2LH_L - (2n-1)/n$ for a star. For a path, first hit either endpoint (mean $i(L-i)$ from position i) and then traverse to the opposite endpoint (mean L^2); averaging gives the formula. The clique and star expressions follow from collecting previously unseen vertices or leaves. Across these 21 checks, the largest absolute standardized discrepancy is 2.66 standard errors; 19 of 21 approximate 95% intervals contain the analytic value. This is a numerical check rather than proof of correctness. Across all 133 graphs, the median relative 95% interval half-width is 1.65%, with maximum 3.76%.

Table 1. Family-balanced errors. Brackets: 95% Monte Carlo intervals for MAPE. First three rows use held-out sizes; LOFO rows use separate family folds over all sizes. The factor is the exponentiated mean absolute log error.

Predictor	MAPE (%)	Factor
Upper bound	2151.1 [2142.5, 2161.7]	7.26
Size-only	75.9 [75.4, 76.4]	1.96
Four statistics	20.1 [19.8, 20.5]	1.20
LOFO size-only	75.4 [75.1, 75.8]	1.94
LOFO four stats.	31.3 [31.1, 31.6]	1.41

Table 2. MAPE (%) by graph family for the four-statistic model. Size-test brackets denote Monte Carlo 95% intervals. LOFO tests the entire omitted family over all seven sizes.

Family	Size test	LOFO
path	13.3 [12.1, 14.6]	14.8
cycle	10.4 [9.4, 11.4]	16.4
star	5.3 [4.8, 6.1]	42.2
clique	10.5 [9.8, 11.3]	58.9
grid	50.1 [48.6, 51.8]	56.9
ER sparse	15.2 [14.7, 15.8]	16.7
ER dense	54.6 [53.6, 55.6]	58.0
random tree	10.0 [9.5, 10.4]	11.1
lollipop	23.3 [21.2, 25.5]	31.9
barbell	8.3 [6.7, 9.9]	6.3

Overall predictive accuracy. Table 1 shows a reduction from 75.9% to 20.1% MAPE when moving from size alone to four statistics; the multiplicative error factor falls from 1.96 to 1.20. The general upper bound is much looser, with 2151% MAPE. That comparison alone would give weak evidence of usefulness; the fitted size baseline provides the more informative comparison. The fitted four-statistic rule, rounded for readability, is

$$\widehat{C}_{\mathrm{pred}} \simeq 0.602\, n^{0.786} m^{0.031} \Delta^{0.696} D^{1.182}. \qquad (3)$$

Full-precision standardized coefficients are saved for reuse. Because features are correlated and the model is regularized, these exponents should not be interpreted as causal scaling laws.

Family dependence and transfer. The family breakdown (Table 2, Figure 1) shows that stars, random trees, and barbells have approximately 5–10% held-out-size error. Paths, cycles, cliques, and sparse ER graphs have 10–15% error,

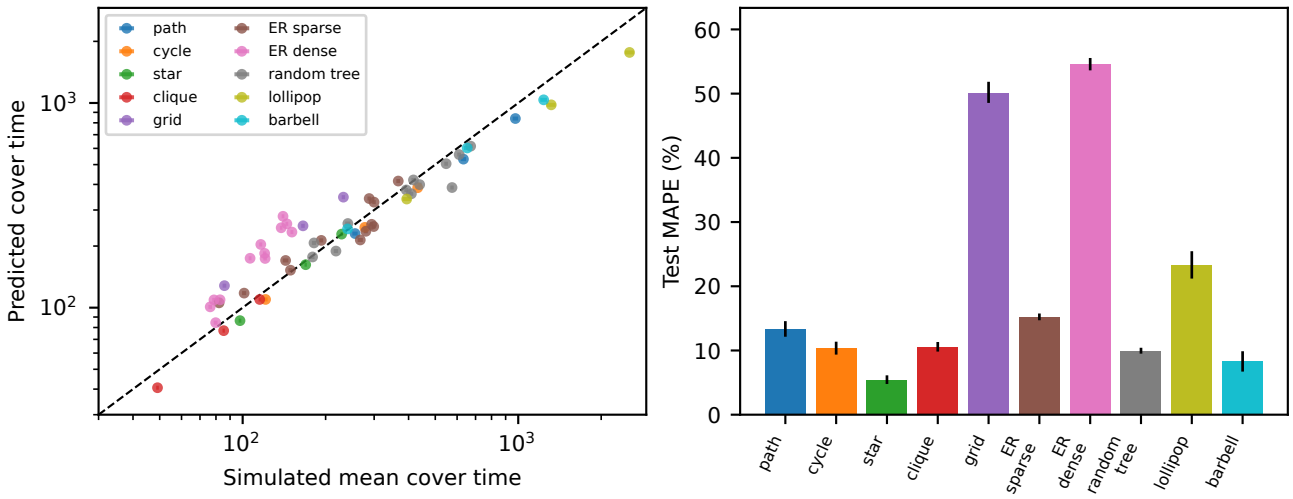


Figure 1. Held-out-size predictions of the four-statistic model. Left: predicted versus simulated cover time, with horizontal 95% Monte Carlo error bars and a dashed equality line. Right: family-wise MAPE; black intervals propagate Monte Carlo uncertainty through refitting. These intervals do not measure variability over a population of graphs.

while lollipops have 23% error. The largest errors occur on grids (50%) and dense ER graphs (55%), both systematically overpredicted on these test graphs. Lollipops are systematically underpredicted. Bottlenecks are therefore not the only difficult case. Nor is adding statistics uniformly better: the size-only baseline's cycle error is 5.6%, versus 10.4% for four statistics.

LOFO MAPE increases to 31.3% [31.1, 31.6], still below its size-only counterpart (75.4%), but stars and cliques now have 42% and 59% error. Thus low error within a represented family does not imply successful transfer to a previously unrepresented family. In contrast, barbell and random-tree LOFO errors remain about 6% and 11%.

4. Discussion and Limitations

Four elementary statistics provide a substantially better approximation than the chosen general bound and a fitted size baseline on this collection. Nevertheless, the family-wise errors are far larger than the simulation uncertainty, indicating model mismatch rather than insufficient walks. The large grid and dense-ER errors are consistent with the limited flexibility of a single multiplicative rule. Diameter and maximum degree do not encode all spatial organization or bottleneck structure. Our experiment does not isolate which omitted property causes each error.

This is a connectivity smoke test, not an exhaustive benchmark: it includes only seven sizes from 8 upward, a few instances per random family, and specific lollipop/barbell proportions. In particular, nothing here establishes a guarantee for every graph with at most 30 vertices, for $n < 8$, for different starting distributions, or for worst-start cover time. Family balancing is a deliberate evaluation choice, not a natural distribution on all connected graphs. Small Monte Carlo intervals should not be mistaken for confidence in out-of-distribution predictions.

The predictor is a rough estimate for represented families and sizes; applications should validate on their own graph class. Predictions are not certified upper bounds. More flexible models remain untested. Code, raw outcomes, seeds, and compilation instructions accompany this report.

References

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